National University of Computer & Emerging Sciences, Karachi
Midterm TWO (Fall-2023)
Nov 06, 2023, 8:30 AM – 9:30 AM

Course Code: CS2001	**Course Name: Data Structures**
Instructor Name: Dr. Jawwad A. Shamsi, Dr. Fahad Sherwani, Mr. Basit Ali, Ms. Sobia Iftikhar, Ms. Mubashra Fayyaz, Mr. Minhal Raza, Dr. Anam Qureshi, Ms. Sumaiyah Zahid, Ms. Sania Urooj, Mr. Farooq Zaidi, Mr. Shahbaz Siddiqui, Ms. Fizza Mansoor, Dr. Farrukh Hasan	
Student Roll No:	**Section No:**

Instructions:

- Return the question paper and make sure to keep it inside your answer sheet.
- Read each question completely before answering it. There are **3 questions and 2 pages**.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- You are **not allowed to write** anything on the question paper (except your ID and group).

Time: 60 minutes. **Max Marks**: 45 Points

Question: 1 [15 points] [CLO 3]
Suppose we have an array arr[] of size n and a block (to be jumped) of size m. Then we search in the indexes arr[0], arr[m], arr[2m]….,arr[km] and so on. Once we find the interval (arr[km] < x < arr[(k+1)m]), we perform a linear search operation from the index km to find the element x.

Let's consider the following array: (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610). The length of the array is 16. The search will find the value of 55 with the following steps assuming that the block size to be jump is 4.

STEP 1: Jump from index 0 to index 4;
STEP 2: Jump from index 4 to index 8;
STEP 3: Jump from index 8 to index 12;
STEP 4: Since the element at index 12 is greater than 55, we will jump back a step to come to index 8.
STEP 5: Perform a linear search from index 8 to get the element 55.

Write the code of the above searching method. Compare this algorithm with the binary search and linear search technique in terms of time complexities. Explain your answers with proper justification(s).

Question: 2 [15 points] [CLO 3]
You are part of a development team working on a collaborative writing platform similar to Google Docs. This platform allows multiple users to work simultaneously on a document. Your task is to implement a feature that tracks and manages undo and redo actions (text only) and keeps the revision history organized and accessible for different team members.
Write code to track Undo & Redo operation between multiple users. You need to store user information as well in the stack to maintain the history.

Sample Main and Output

```
int main() {
    int historyCapacity = 100;
   DocumentEditor editor(historyCapacity);
   User user1("User1");
```

```
    User user2("User2");

    editor.push(user1, "This is the first edit by User1. ");
    editor.push(user2, "User2 made an addition.");
    editor.showDocument();
    editor.undo();
    editor.showDocument();
    editor.redo();
    editor.showDocument();
}
```

Sample Output :
This is the first edit by User1. User2 made an addition.
This is the first edit by User1.
This is the first edit by User1. User2 made an addition.

Question: 3 [15 points] [CLO 3]
Consider an BST after a sequence of insertions and deletions. The in-order/in-fix traversal of the currently stored nodes in the BST is 10, 20, 30, 40, 50 & 60.

1. Check whether it is an AVL tree or not? If not, identify the node that violates the condition and provide the corrected AVL tree. (Note: No need to write code. Just identify if any node is violating the condition and identify it.)
2. Perform the following operation after correcting the AVL tree in part (1).
    a. Insert 25
    b. Delete 40
    c. Insert 35
    d. Delete 10

Note: For every operation, please show each step clearly i.e. calculation of balance factor, rotation, etc.